

CRIME BUSTERS

DESCRIPTION: A team of two students will be asked to solve a “crime” by identifying common household white powders, evaluating fingerprints, and completing paper chromatography.

SAFETY PARAMETERS:

Safety Precautions MUST be used:

- Participants dress for the event including closed-toed shoes, long pants, and long sleeve shirt
- Participants must wear safety goggles and lab coat or apron (lab coat may replace long sleeve shirt)
- Shoulder length or longer hair must be tied back
- Taste testing is strictly prohibited

COMPETITORS BRING/NEED:

- Pencils/Pens
- Powders Properties Chart
- Goggles
- Lab Coat or Apron
- Closed-toed shoes must be worn
- Gloves are optional

NUMBER OF PARTICIPANTS: Maximum of 3

IMPOUND: NO

THE COMPETITION:

1. Students will be given a scenario and clues to solve a “crime”
2. There will be 4 suspects with associated evidence to analyze.
 - a. The competition has 4 parts and may be run as stations:
 - i. Powder Identification: Teams may be asked to identify the following powders in order to solve a simulated crime. Powders will be limited to granulated sugar, flour, cornstarch, baking soda, calcium carbonate (chalk), powdered Alka Seltzer®, plaster of paris, and sand.
 1. Event supervisor will provide powder samples from the “crime”, popsicle sticks or scoops for stirring and scooping powders, hand lens, paper towels, iodine for starch test, white vinegar, droppers or transfer pipettes for liquids, distilled water, and container to test powders in (ex. Well plate, plastic containers, plastic plate)
 2. Participants will create a Powder Properties chart describing the powders’ reactions to the above materials before attending the tournament. 2 copies of this chart should be brought to the tournament. One copy of the chart and answer sheet will be collected at the end of the event.
 3. The samples of powder will be one single powder and not a mixture.
 4. There will be only one powder per suspect to analyze.
 5. Tasting, touching or feeling of the substances or unsafe handling of the equipment is NOT allowed.
 6. No extra powder is provided, so students should practice with small amounts.
 - ii. Fingerprint Identification: Teams will be given a set of fingerprints from several suspects and from the “crime” scene. They will be asked to match the fingerprints found at the scene and identify types of fingerprint patterns such as loops, arches, and whorls.

- iii. Paper Chromatography: Teams will need to perform simple chromatography on markers/pens belonging to suspects in order to match the evidence from the “crime” scene. There will be no more than 1 sample from each suspect. Students will be provided with water, beakers or plastic cups, chromatography or filter paper strips, and various markers/pens to match the evidence from the “crime” scene. No calculations will be performed.
- iv. Case Report: Students analyze the data collected from the three types of evidence and make a conclusion as to who committed the crime and the evidence that supports their conclusion.
- b. Teams will clean stations and dispose of waste as directed by the event supervisor.
- c. Teams must leave their Powders Properties Chart with the event supervisor. The quality of the chart will be used as the first tie-breaker.

RELATED FLORIDA STANDARDS:

- SC.5.N.1.1 Define a problem, use appropriate reference materials to support scientific understanding, plan and carry out scientific investigations of various types such as: systematic observations, experiments requiring the identification of variable, collecting and organizing data, interpreting data in charts, tables, and graphics; analyze information; make predictions; and defend conclusions.
- SC.4.N.1.6 Keep records that describe observations made, carefully distinguishing actual observations from ideas and inferences about the observations.
- SC.3.N.1.3 Keep records as appropriate, such as pictorial, written, or simple charts and graphs, of investigations conducted.
- SC.3.N.1.6 Infer based on observation
- SC.4.N.1.4 Attempt reasonable answers to scientific questions and cite evidence in support.
- SC.5.P.8.1 Compare and contrast the basic properties of solids, liquids, and gases, such as mass, volume, color, texture, and temperature.
- SC.5.P.8.2 Investigate and identify materials that will dissolve in water and those that will not and identify the conditions that will speed up or slow down the dissolving process.

SCORING:

- High Score wins
- The score will be based on the following formula:
 - Powder identification - 50 points
 - Fingerprint analysis - 20 points
 - Chromotography - 20 points
 - Final Analysis - 25 points
- Tie breakers will be:
 - First tie-breaker - Quality of Powder Properties Table
 - Second tie-breaker - Quality of Final Analysis
- Teams who do not have proper safety equipment cannot compete. They may leave to find someone to lend them the equipment but will not receive any time back to make up for lost time. The Event Supervisor may provide safety equipment, but there will be a 10% penalty given. This penalty is taken off the overall score that the team gets.
- A penalty of 10% will be given if the lab station is not cleaned up as instructed by the event supervisor. This penalty is taken off the overall score that the team gets.

Crime Busters SAMPLE Answer Sheet (May be adapted by Event Supervisor)

School: _____ Team Number: _____

Team Member 1: _____ Team Member 2: _____

Points for Part 1	_____/50
Points for Part 2	_____/20
Points for Part 3	_____/20
Points for Part 4	_____/25
Total	_____/115
Penalty deduction	_____
Final Score	_____/115
Tiebreaker?	_____/10

Notes/Data Collection:

[illegible]

Instructions: Complete the analysis using observations - use these sheets for answers. No additional paper allowed.

Part 1: Powder Evidence:

	Evidence	Analysis of the Evidence (Answer)	Score
1a	What is the identity of the powder found on Suspect 1?		___/4
1b	How did you identify it?		___/5
2a	What is the identity of the powder found on Suspect 2?		___/4
2b	How did you identify it?		___/5
3a	What is the identity of the powder found on Suspect 3?		___/4
3b	How did you identify it?		___/5
4a	What is the identity of the powder found on Suspect 4?		___/4
4b	How did you identify it?		___/5
5a	What happened when you tested the evidence powder found at the scene?		___/5
5b	What powder was found at the scene?		___/5
5c	Which suspect's mystery powder is the same as the powder at the crime scene?		___/4
Total points for Section 1			___/50

Part 2: Fingerprint Analysis:

	Evidence	Analysis of the Evidence (Answer)	Score
1a	Type of fingerprint pattern of suspect 1		___/3
1b	Type of fingerprint pattern of suspect 1		___/3

1c	Type of fingerprint pattern of suspect 3		___/3
1d	Type of fingerprint pattern of suspect 4		___/3
1e	Type of fingerprints pattern found at the crime scene		___/3
1f	Whose fingerprints were like those at the scene?		___/5
Total points for Section 2			___/20

Part 3: Paper Chromatography:

Complete a chromatograph according to the directions at the lab station.

3a	Paperclip or staple your chromatography below. Points will be awarded for the quality of separation.		___/15
3f	Points awarded for: Marking the baseline (2.5) Marking the solvent front (2.5)		___/5
Total points for Section 3			___/20

Part 4: Case Report:

Instructions: Using the evidence you analyzed, draw a conclusion as to what happened about the event. Please consider evidence that supports the guilt of certain suspect(s) and evidence that supports the innocence of the other suspects. This section will be used as the second tiebreaker, if needed.

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